

ARNOB RIZWAN AHMAD

AI / LLM Engineer • Agentic Systems • RAG & Evaluation • Python Backend • Cloud
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PROFILE

AI/LLM engineer and Software Engineering graduate of Universiti Teknologi Malaysia (July 2026). Came to AI from production full-stack and ERP work, and build LLM systems the same way — agentic workflows, RAG over FAISS and pgvector, and evaluation harnesses with blinded LLM-judge scoring and execution guardrails, on the FastAPI, PostgreSQL and Kubernetes infrastructure they run on. Went deep enough on the systems side to certify as an AMD ROCm Associate. Seeking AI/ML engineering roles where evaluation and reliability matter as much as model choice.

EDUCATION

Universiti Teknologi Malaysia (UTM) — Johor Bahru

Oct 2022 – July 2026

Bachelor of Software Engineering (Computer Science) — Graduated July 2026

PROFESSIONAL EXPERIENCE

Agentic AI Engineer — BLOOMTECH, Inc. — Remote (Part-time)

Aug 2026 – Present

Agentic AI systems and LLM-driven tooling

- Build agentic AI workflows in Python — multi-step LLM planning, tool and API integration, and structured output handling.
- Design prompt and evaluation loops that measure and improve agent reliability across task categories.

Enterprise Resource Planning Developer — Hirdaramani Bangladesh — Hybrid (Full-time)

Jul 2026 – Present

ERP systems for a multinational apparel manufacturing group

- Build and maintain ERP modules supporting production, inventory and order-management workflows across factory operations.
- Write and tune SQL against large operational datasets; build reports and data integrations between ERP and internal systems.
- Work with operations and IT stakeholders to translate process requirements into shipped features.

Application Developer (Freelance) — Moien Delivery — Luxembourg (Remote)

Mar 2026 – Jul 2026

Hybrid logistics platform serving customers, restaurants and drivers — live on Google Play & Apple App Store

- Built the Angular web front-end and a cross-platform React Native mobile app on a Nuxt.js backend, delivering features for three distinct user roles.
- Implemented real-time order tracking, driver dispatch and delivery workflow automation over REST APIs and WebSocket event streams.
- Owned features from spec through QA to production deployment on weekly release cycles with international stakeholders.

Machine Learning Engineer (Contract via Mercor) — AI News HUD — USA (Remote)

Oct 2025 – Mar 2026

US-based AI-powered news intelligence platform

- Built and evaluated production PyTorch models for real-time news classification, summarization and personalized recommendations.
- Designed evaluation pipelines and ran iterative experiments on live traffic, raising model accuracy to **78.9%** and cutting inference latency by **20%**.
- Took ML components from prototype to deployed feature in the platform's core intelligence layer.

Freelance & Contract Engineering

Sep 2024 – Jul 2026

Concurrent client and programme work delivered alongside study

- **Framic** (remote, Aug 2025 – Jul 2026) — founding backend engineer at an AI storage startup; designed backend services from scratch, owned architecture and core API design, and led AI search and real-time file indexing.
- **MOMS Mobile Oil Change** (USA, Apr – Jul 2026) — production Next.js site, live at moms oilchange.com: rebuilt navigation and three product pages, led an SEO overhaul, and ran a site-wide security and dependency audit.
- **Headstarter** (USA, Sep 2024 – Mar 2025) — competitive 6-month software engineering residency; shipped 14+ ML and full-stack projects, trained 5 neural networks, and cut image size and build time on a Docker-based GitHub Actions pipeline.

SELECTED PROJECTS

School Care App — 40-School Deployment — Sylhet, Bangladesh

Mar 2026

- Delivered a full-stack school management system adopted by 40 schools across Sylhet; in closed testing ahead of Google Play launch.
- Built a React admin dashboard and a React Native app for parents and teachers, with Cloud Functions fanning out role-targeted FCM push notifications by school, role, and class.

ShopLocalr — Local Commerce Mobile App — Malaysia

Feb 2026

- Shipped a 33,000-line Flutter local-business discovery app with customer, business, and admin roles — QR deal redemption with checkout PINs, Gemini-backed recommendations, and offline-first Hive caching on a Firebase backend.

Autonomous ML Agent — LLM-Planned AutoML Pipeline (GitHub)

- Built an AutoML system where an LLM planner proposes modeling strategies under a time budget, Optuna tunes XGBoost/LightGBM/scikit-learn models, ensembles are built, and SHAP explains the winner.
- Shipped with model cards, a FastAPI inference service, 5 pytest suites, and CI/CD publishing to Docker Hub.

Self-Improving Prompt Optimization API — LLM Evals Infrastructure (GitHub)

- Built CI/CD for prompts: a semver-versioned prompt registry where each improvement cycle runs failure analysis, generates candidates, A/B-evaluates them with a blinded LLM judge, and auto-promotes winners per YAML rules.
- Multi-dimensional scoring across OpenAI and Anthropic providers with FastAPI, Postgres, and Redis.

CrisisCohort — Multi-Agent AI Climate Resilience Platform (GitHub)

- Architected a climate-resilience platform for smallholder farmers with 10+ specialized AWS Bedrock agents (water optimization, climate migration, grant matching, disease diagnostics) across 15,000+ lines of Lambda.
- Designed an AWS architecture spanning Lambda, Bedrock, DynamoDB, CloudFront, API Gateway, and Cognito, plus an on-chain water-rights trading contract in Solidity.

Terminal Coding Agent — Sandboxed CLI Agent (GitHub)

- Built a pip-installable CLI coding agent that plans edits via OpenAI/Anthropic, applies them through a diff manager, and executes inside E2B Firecracker microVMs.
- Added guardrails scanning for six secret types (AWS, GitHub, Slack keys) and 5 pytest suites.

TECHNICAL SKILLS

ML / AI: PyTorch, scikit-learn, XGBoost, LightGBM, Optuna, SHAP; LLM agents, RAG (FAISS, pgvector), LLM evaluation & blinded judges, prompt engineering, AWS Bedrock, Gemini

GPU / HPC: AMD ROCm, HIP, CUDA→HIP porting, rocprofv3 profiling, mixed-precision (FP16/BF16) GEMM, kernel tuning

Languages: Python, SQL, TypeScript, JavaScript, C#, Java, Dart

Backend: FastAPI, Django + DRF, Node.js, Express, NestJS, ASP.NET Core, Celery, REST APIs, WebSockets

Frontend & Mobile: React, Next.js, Angular, React Native, Flutter

Cloud & Data: AWS, Microsoft Azure, GCP, Docker, Kubernetes, Linux, CI/CD (GitHub Actions, Jenkins); PostgreSQL, MySQL, SQL Server, MongoDB, Redis, ERP modules and SQL tuning on large operational datasets

CERTIFICATIONS

- **AMD — ROCm Certified Associate** (Aug 2026)
- **AWS** — Building Language Models on AWS; Customizing & Evaluating LLMs Using Amazon SageMaker JumpStart; Amazon SageMaker JumpStart Foundations; Foundations of Prompt Engineering (Mar 2025)
- **IBM** — Full Stack Software Developer Professional Certificate (Aug 2024)
- **Microsoft Azure AI** — Azure OpenAI Service, Natural Language Processing, Document Intelligence (Jun 2024)